



CERTIFIED MATERIAL TEST REPORT
CERTIFICATE OF COMPLIANCE
CERTIFICATE # 246439
REVISION 0

Tube Forgings of America, Inc.
5200 N.W. Front Avenue
Portland, OR 97210
ISO 9001:2015 Certified
4/15/2026

Sold To	Dodson Valco, Inc 2750 Holmes Rd. Houston, TX 77051	Ship To	Dodson Valco, Inc 2750 Holmes Rd. Houston, TX 77051
Customer ID	9401		
TFA Sales Order	328401\12-1		
Purchase Order	440775-00		

TFA Part ID	52066SG	Heat Code	A3207B
Description	16 XH LR 90 Y-65 SEGM	Grade	Y65
Quantity	20	Raw Material Heat	68645
Customer Part ID		Process	Y65-7A
		Mill	TAMSA

Austenitized @:	1650°	Tempered @:	1200°	Stress Relieved @:		Medium:	WATER
Held For:	60 mins/min	Held For:	60 mins/min	Held For:	mins/min		

Heat Analysis Chemistry

C	Mn	P	S	Si	Mo	Cr	Ni	Cu	V	Cb	B	Ti	Al	N	Co	Ca	C.E.
0.17	1.08	0.012	0.001	0.23	0.03	0.06	0.06	0.15	0.04	0.02	0.0000	0.02	0.03				0.39

Product Analysis Chemistry

C	Mn	P	S	Si	Mo	Cr	Ni	Cu	V	Cb	B	Ti	Al	N	Co	Ca	C.E.
0.18	1.07	0.011	0.001	0.24	0.03	0.06	0.07	0.15	0.04	0.02	<0.0010	0.02	0.03				0.40

Tensile Test Results

Test	Name	Yield	Tensile	% Elong	R/A %	Gage Length	HBW	In-Process Test 1 Test 2	Tensile Orientation	Test Bar Type & Size
2	BASE	68700 (psi)	86600 (psi)	31.90	75.40	4D	182		TRANSVERSE	ROUND (.252" Diameter)

Charpy Impact Test Results

		Impact Energy				Lateral Expansion (Mils)			% Shear				Test Temp.	Size (mm)	Charpy Orientation
		1	2	3	Avg	1	2	3	1	2	3	Avg			
BASE	FtLbs	186.00	194.00	145.00	175.00				95.00	100.00	100.00		+20F	10X7.5	TRANSVERSE

Notes/Comments

1. THIS MATERIAL WAS AUSTENITIZED AT 1650F, WATER QUENCHED, AND TEMPERED AT 1200F. 2. THIS MATERIAL IS OF SEAMLESS MANUFACTURE.

*UB = Unbroken Test

Heat: A3207B Grade: Y65

Page 1 of 2


Scot Marshall, Manager Quality Assurance

The recording of fictitious entries on this document and or its fraudulent misuse may be punishable as a felony under Federal Statute.



CERTIFIED MATERIAL TEST REPORT
CERTIFICATE OF COMPLIANCE
CERTIFICATE # 246439
REVISION 0

Tube Forgings of America, Inc.
5200 N.W. Front Avenue
Portland, OR 97210
ISO 9001:2015 Certified
4/15/2026

Sold To	Dodson Valco, Inc 2750 Holmes Rd. Houston, TX 77051	Ship To	Dodson Valco, Inc 2750 Holmes Rd. Houston, TX 77051
Customer ID	9401		
TFA Sales Order	328401\12-1		
Purchase Order	440775-00		

TFA Part ID	52066SG	Heat Code	A3207B
Description	16 XH LR 90 Y-65 SEGM	Grade	Y65
Quantity	20	Raw Material Heat	68645
Customer Part ID		Process	Y65-7A
		Mill	TAMSA

- 1) Fittings comply with the requirements of MSS-SP-75 - 2025.
- 2) Product complies with ANSI/NACE MR0175-2021/ISO 15156-2:2020, Annex A, section A.2.1.2 and A.2.1.3, and ANSI/NACE MR0103/ISO 17945:2015, section 13.1.
- 3) Product is manufactured in an ISO 9001 certified facility: Certificate #30248.
- 4) Fittings under 16 NPS do not require notch toughness testing unless required by MSS SP-75 or supplementary requirements SR-6 or SR-24 are invoked per MSS SP-75.
- 5) Fitting schedules XH and XXH are synonymous with XS and XXS.

Elbows identified as SEGMENTABLE are certified to be Field Segmentable, maintain an inside diameter of no less than 98% of the specified nominal inside diameter, and be capable of receiving a pigging device suited to travel through a fitting with an inside diameter not less than 98% of the specified nominal inside diameter (taper bored fittings are excluded from pigging compliance).

Tee runs are straight bored, maintain an inside diameter of no less than 98% of the specified nominal inside diameter, and are capable of receiving a pigging device suited to travel through a fitting with an inside diameter not less than 98% of the specified nominal inside diameter (tee runs ordered as taper bored are excluded from pigging compliance).

Material Formed/Forged At: Tube Forgings of
America, Inc, Portland OR

Material Heat Treated At: Met-Tek,
Clackamas OR

TFA FITTINGS ARE MANUFACTURED IN PORTLAND, OR, USA. FOR THIS HEAT CODE, RAW MATERIAL SOURCE IS MEXICO

*UB = Unbroken Test

Heat: A3207B Grade: Y65

Page 2 of 2

Scot Marshall, Manager Quality Assurance

The recording of fictitious entries on this document and or its fraudulent misuse may be punishable as a felony under Federal Statute.